Lesson 10: Bootstrap Confidence Intervals

BSTA 551: Statistical Inference

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Lesson 10: Bootstrap Confidence Intervals

Review: Bootstrap Basics (Lesson 9)

The Bootstrap Principle

1. **The sample represents the population** — resample from the sample
2. **Sampling with replacement** creates variability
3. **Bootstrap distribution approximates sampling distribution**

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The Bootstrap Principle

1. **The sample represents the population** — resample from the sample
2. **Sampling with replacement** creates variability
3. **Bootstrap distribution approximates sampling distribution**

Key algorithm:

1. Resample n observations **with replacement** from the original sample
2. Compute statistic $\hat{\theta}^*$ from bootstrap sample
3. Repeat B times (typically $B = 10,000$)
4. Analyze the distribution of $\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*$

Quick Review: What Bootstrap Can Estimate

Can estimate:

- Standard error of $\hat{\theta}$
- Bias of $\hat{\theta}$
- Shape/skewness of sampling distribution
- Confidence intervals

Cannot improve:

- Point estimates (centered at $\hat{\theta}$, not θ)
- A bad/unrepresentative sample

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Cannot improve:

- Point estimates (centered at $\hat{\theta}$, not θ)
- A bad/unrepresentative sample

Today's Goals:

1. Compute bootstrap standard errors
2. Construct bootstrap confidence intervals (percentile and t methods)
3. Compare methods and know when to use each
4. Apply bootstrap to complex statistics (trimmed means, ratios)

Running Example: Drug Response Times

```r
response_times <- c(8, 12, 15, 7, 22, 45, 11, 9, 14, 18, 10, 13)
n <- length(response_times)

cat("n =", n, "\n")
```

```
n = 12
```

```r
cat("Mean:", round(mean(response_times), 2), "days\n")
```

```
Mean: 15.33 days
```

```r
cat("Median:", median(response_times), "days\n")
```

```
Median: 12.5 days
```

```r
cat("SD:", round(sd(response_times), 2), "days")
```

```
SD: 10.27 days
```

Note the outlier at 45 days — this motivates robust methods!

Bootstrap Standard Errors

Bootstrap Standard Error

Definition

The **bootstrap standard error** of a statistic $\hat{\theta}$ is the standard deviation of its bootstrap distribution:

$$SE_{boot}(\hat{\theta}) = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}(\hat{\theta}_b^* - \bar{\hat{\theta}}^*)^2}$$

where $\bar{\hat{\theta}}^* = \frac{1}{B}\sum_{b=1}^{B}\hat{\theta}_b^*$

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where $\bar{\hat{\theta}}^* = \frac{1}{B}\sum_{b=1}^{B}\hat{\theta}_b^*$

This gives us an **empirical estimate** of the standard error without assuming any specific distribution!

Medical Example: Blood Pressure Reduction

A pilot study measures systolic blood pressure reduction (mmHg) after 8 weeks on a new antihypertensive:

```
bp_reduction <- c(12, 8, 15, 22, 5, 18, 11, 14, 9, 25, 7, 16, 13, 10, 19)
n_bp <- length(bp_reduction)

cat("Sample size:", n_bp, "\n")
```

```
Sample size: 15
```

```
cat("Mean reduction:", round(mean(bp_reduction), 2), "mmHg\n")
```

```
Mean reduction: 13.6 mmHg
```

```
cat("Sample SD:", round(sd(bp_reduction), 2), "mmHg\n")
```

```
Sample SD: 5.67 mmHg
```

```
cat("Classical SE:", round(sd(bp_reduction)/sqrt(n_bp), 2), "mmHg")
```

```
Classical SE: 1.46 mmHg
```

Bootstrap SE for Blood Pressure

```r
set.seed(551)
B <- 10000
boot_means_bp <- replicate(B, {
  boot_sample <- sample(bp_reduction, n_bp, replace = TRUE)
  mean(boot_sample)
})

# Compare standard errors
classical_se <- sd(bp_reduction) / sqrt(n_bp)
boot_se <- sd(boot_means_bp)

cat("Classical SE (s/√n):", round(classical_se, 3), "\n")
```

```
Classical SE (s/√n): 1.463
```

```r
cat("Bootstrap SE:", round(boot_se, 3), "\n")
```

```
Bootstrap SE: 1.41
```

```r
cat("Ratio:", round(boot_se / classical_se, 3))
```

```
Ratio: 0.964
```

Bootstrap SE for Blood Pressure

```r
set.seed(551)
B <- 10000
boot_means_bp <- replicate(B, {
  boot_sample <- sample(bp_reduction, n_bp, replace = TRUE)
  mean(boot_sample)
})

# Compare standard errors
classical_se <- sd(bp_reduction) / sqrt(n_bp)
boot_se <- sd(boot_means_bp)

cat("Classical SE (s/√n):", round(classical_se, 3), "\n")
```

```
Classical SE (s/√n): 1.463
```

```r
cat("Bootstrap SE:", round(boot_se, 3), "\n")
```

```
Bootstrap SE: 1.41
```

```r
cat("Ratio:", round(boot_se / classical_se, 3))
```

```
Ratio: 0.964
```

The bootstrap SE is very close to the classical SE when assumptions are met!

When Bootstrap SE Shines: Trimmed Mean

For **trimmed means** (robust to outliers), we don't have simple SE formulas.

```r
# Data with potential outlier
response_times <- c(8, 12, 15, 7, 22, 45, 11, 9, 14, 18, 10, 13)

# Bootstrap for 10% trimmed mean
set.seed(551)
boot_trimmed <- replicate(10000, {
  x <- sample(response_times, length(response_times), replace = TRUE)
  mean(x, trim = 0.10)  # Trim 10% from each end
})

cat("Sample mean:", round(mean(response_times), 2), "\n")
```

```
Sample mean: 15.33
```

```r
cat("10% Trimmed mean:", round(mean(response_times, trim = 0.10), 2), "\n")
```

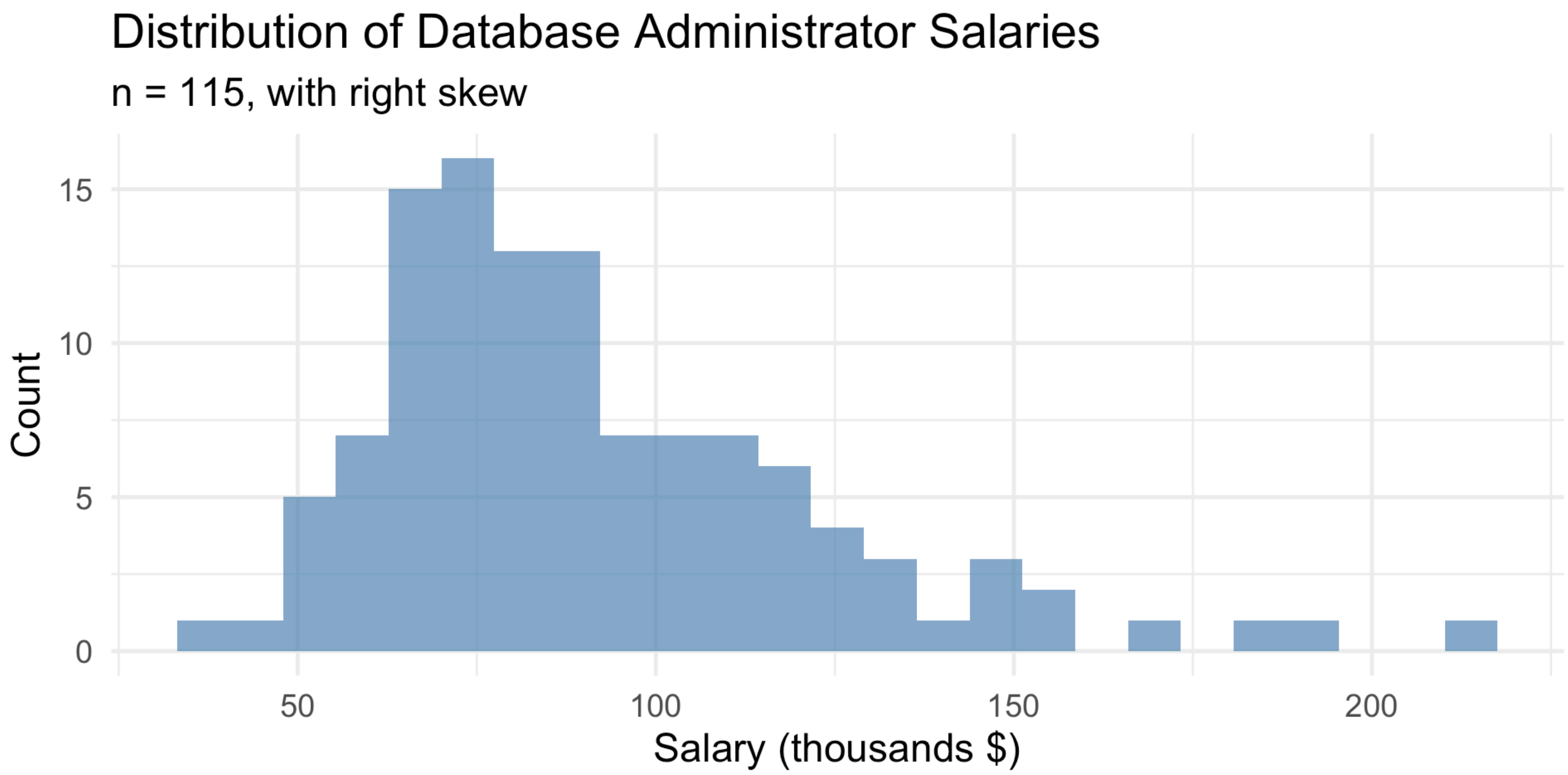
```
10% Trimmed mean: 13.2
```

```r
cat("Bootstrap SE of trimmed mean:", round(sd(boot_trimmed), 2))
```

```
Bootstrap SE of trimmed mean: 2.67
```

Example: Database Administrator Salaries (Devore 8.18)

▶ Code



Bootstrap for Trimmed Mean Salary

```r
# 10% trimmed mean
xbar_tr <- mean(salaries, trim = 0.10)

# Bootstrap
set.seed(551)
B <- 10000
boot_trimmed_sal <- replicate(B, {
  x <- sample(salaries, n_sal, replace = TRUE)
  mean(x, trim = 0.10)
})

cat("10% Trimmed mean:", round(xbar_tr, 0), "\n")
```

```
10% Trimmed mean: 88430
```

```r
cat("Bootstrap SE:", round(sd(boot_trimmed_sal), 0), "\n")
```

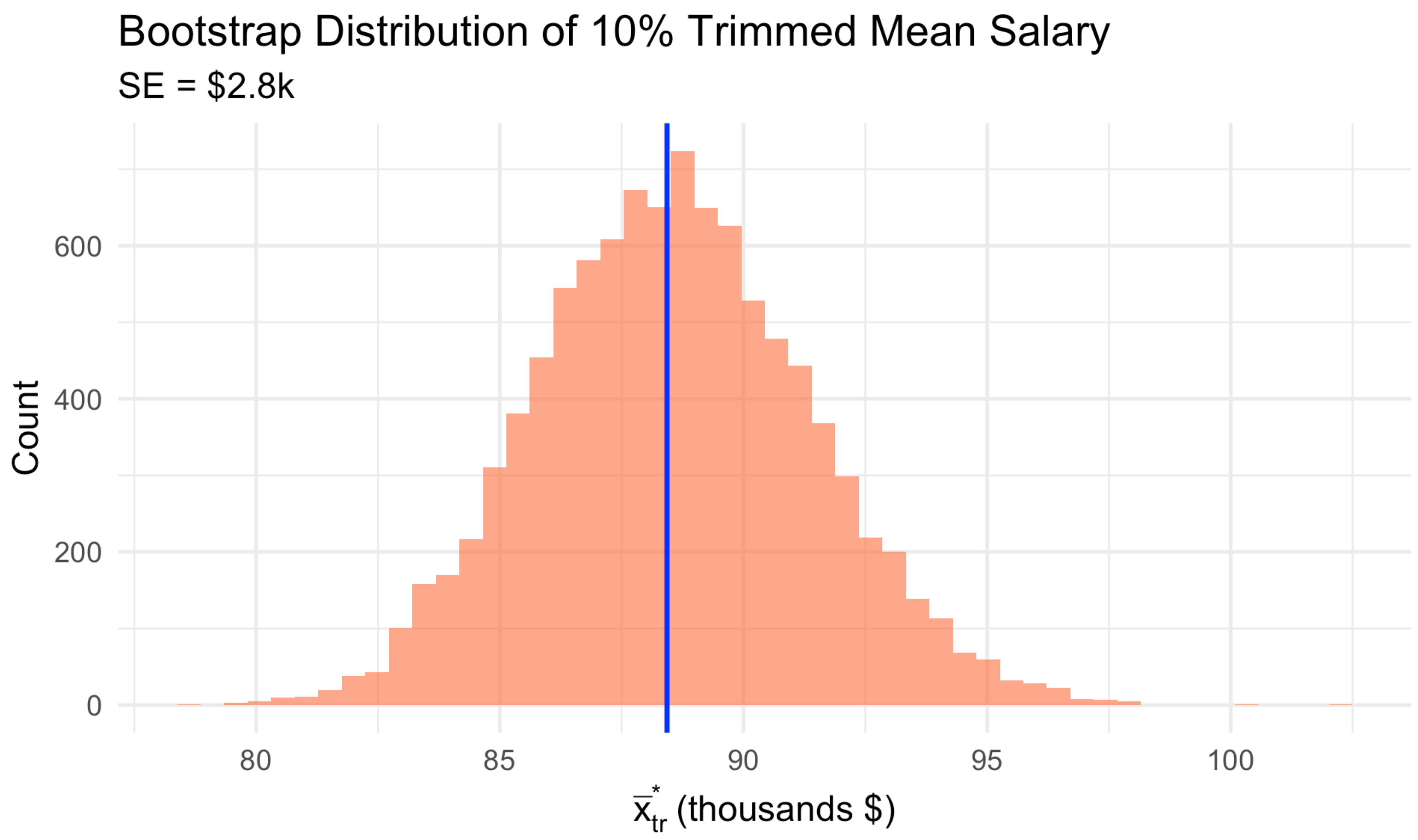
```
Bootstrap SE: 2808
```

```r
cat("Bootstrap mean of trimmed means:", round(mean(boot_trimmed_sal), 0))
```

```
Bootstrap mean of trimmed means: 88561
```

Visualizing the Bootstrap Distribution

▶ Code



The bootstrap distribution appears approximately normal — good for t-based intervals!

Bias in Bootstrap Distributions (Chihara 5.6)

Bootstrap Bias Estimate

The **bootstrap estimate of bias** is:

$$\widehat{\text{Bias}} = \bar{\hat{\theta}}^* - \hat{\theta}$$

where $\bar{\hat{\theta}}^*$ is the mean of the bootstrap distribution and $\hat{\theta}$ is the original sample statistic.

```r
# Check bias for trimmed mean
boot_bias <- mean(boot_trimmed_sal) - xbar_tr
cat("Bootstrap bias estimate:", round(boot_bias, 0), "\n")
```

```
Bootstrap bias estimate: 131
```

```r
cat("Bias as % of estimate:", round(100 * boot_bias / xbar_tr, 2), "%")
```

```
Bias as % of estimate: 0.15 %
```

Bias in Bootstrap Distributions (Chihara 5.6)

Bootstrap Bias Estimate

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```
# Check bias for trimmed mean
boot_bias <- mean(boot_trimmed_sal) - xbar_tr
cat("Bootstrap bias estimate:", round(boot_bias, 0), "\n")
```

```
Bootstrap bias estimate: 131
```

```
cat("Bias as % of estimate:", round(100 * boot_bias / xbar_tr, 2), "%")
```

```
Bias as % of estimate: 0.15 %
```

Small bias suggests the bootstrap distribution is well-centered!

Bootstrap Confidence Intervals

Types of Bootstrap Confidence Intervals

Several methods exist for constructing bootstrap CIs:

1. **Bootstrap Percentile Interval** — Simple, intuitive
2. **Bootstrap t Interval** — Better theoretical properties
3. **BCa Interval** — Bias-corrected and accelerated (advanced)

Types of Bootstrap Confidence Intervals

Several methods exist for constructing bootstrap CIs:

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Practical Guidance

- **Percentile interval:** Good when bootstrap distribution is symmetric, low bias
- **Bootstrap t interval:** Better for small samples, handles skewness
- Both require the bootstrap distribution to be a reasonable approximation

Bootstrap Percentile Interval (Devore 8.5)

Definition

A **95% bootstrap percentile interval** for parameter θ has endpoints equal to the **2.5th and 97.5th percentiles** of the bootstrap distribution.

More generally, a $100(1-\alpha)\%$ bootstrap percentile interval uses the $\alpha/2$ and $1-\alpha/2$ quantiles:

$$\left(\hat{\theta}^*_{\alpha/2},\ \hat{\theta}^*_{1-\alpha/2}\right)$$

Bootstrap Percentile Interval (Devore 8.5)

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$$\left(\hat{\theta}^*_{\alpha/2},\ \hat{\theta}^*_{1-\alpha/2}\right)$$

Intuition: If 95% of bootstrap statistics fall between L and U, then (L, U) is a plausible range for the true parameter.

Computing Percentile Intervals in R

```r
# Blood pressure reduction data
set.seed(551)
bp_reduction <- c(12, 8, 15, 22, 5, 18, 11, 14, 9, 25, 7, 16, 13, 10, 19)

# Bootstrap
boot_means_bp <- replicate(10000, mean(sample(bp_reduction, length(bp_reduction), replace =

# 95% Percentile interval
percentile_ci <- quantile(boot_means_bp, probs = c(0.025, 0.975))

cat("Sample mean:", round(mean(bp_reduction), 2), "mmHg\n")
```

```
Sample mean: 13.6 mmHg
```

```r
cat("95% Bootstrap Percentile CI:", round(percentile_ci, 2), "mmHg\n")
```

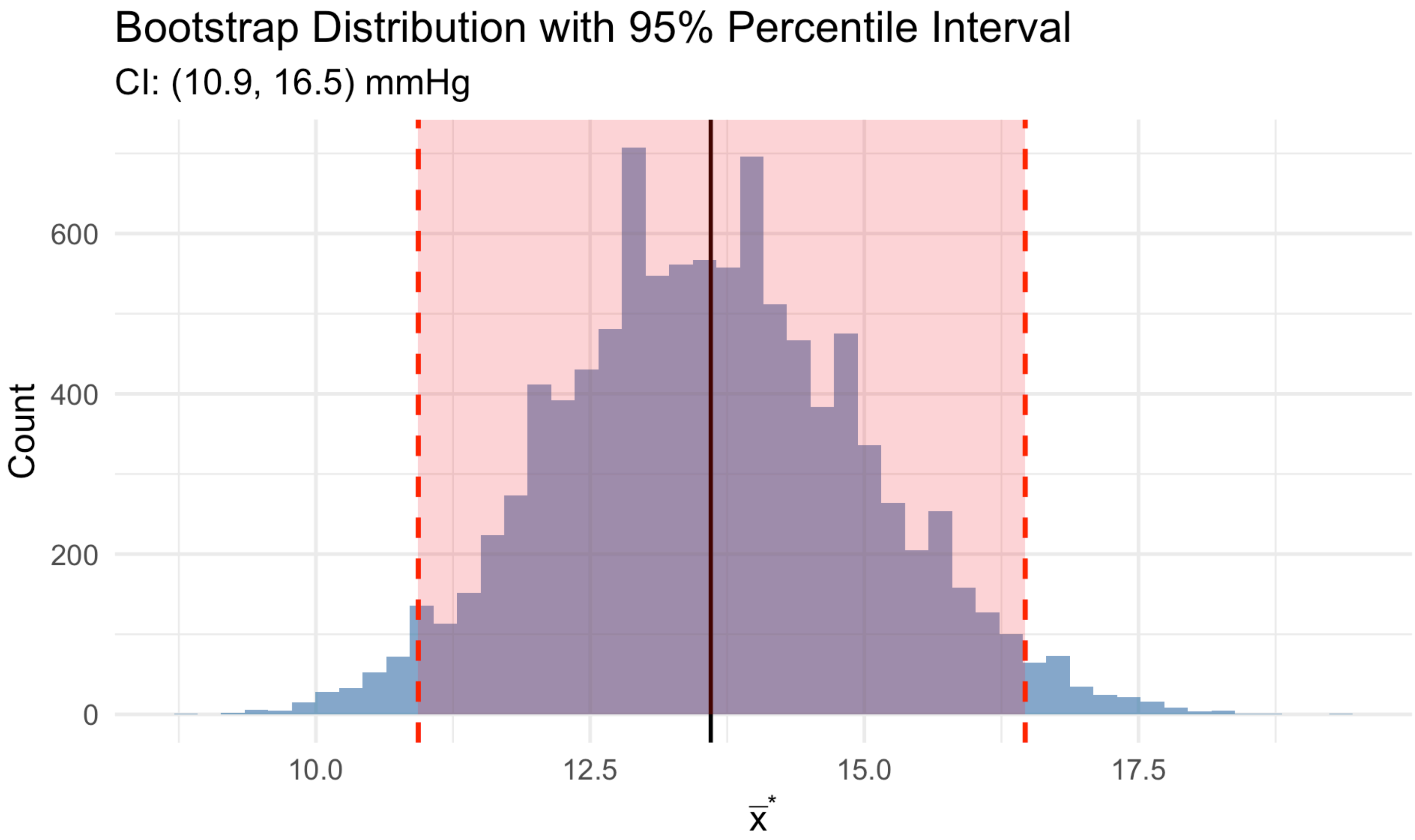
```
95% Bootstrap Percentile CI: 10.93 16.47 mmHg
```

```r
# Compare to t-interval
t_ci <- t.test(bp_reduction)$conf.int
cat("95% t-interval:", round(t_ci, 2), "mmHg")
```

```
95% t-interval: 10.46 16.74 mmHg
```

Visualizing the Percentile Interval

▶ Code



Medical Example: Median Survival Time

For **median** survival, we don't have simple formulas for CIs. Bootstrap is ideal!

```r
# Survival times (months) for 20 cancer patients
survival_times <- c(4, 8, 12, 15, 18, 6, 9, 24, 36, 11,
                    7, 14, 21, 10, 13, 5, 17, 22, 8, 19)

# Bootstrap for median
set.seed(551)
boot_medians <- replicate(10000, {
  median(sample(survival_times, length(survival_times), replace = TRUE))
})

# Results
cat("Sample median:", median(survival_times), "months\n")
```

```
Sample median: 12.5 months
```

```r
cat("Bootstrap SE:", round(sd(boot_medians), 2), "months\n")
```

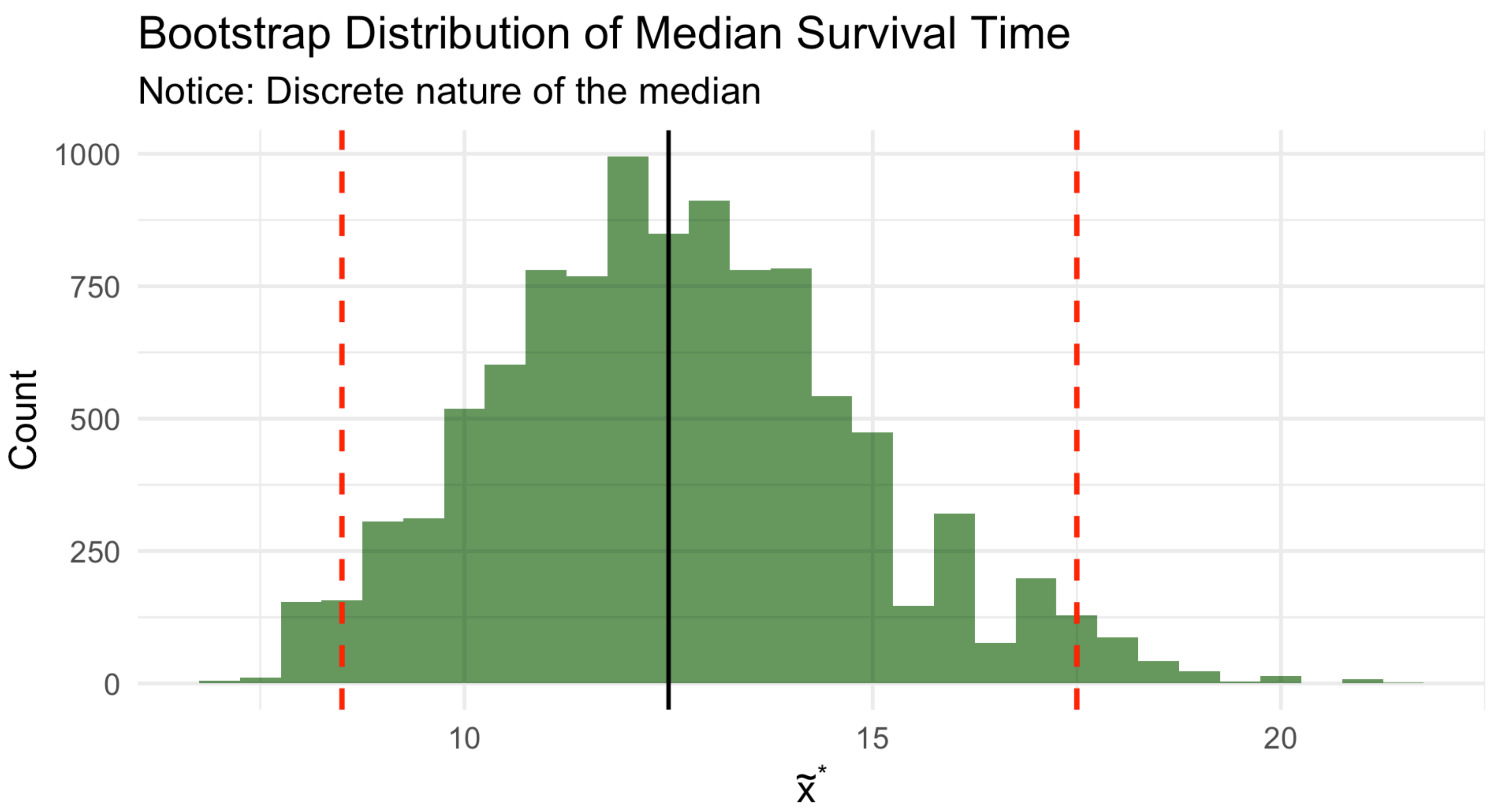
```
Bootstrap SE: 2.22 months
```

```r
cat("95% Percentile CI:", quantile(boot_medians, c(0.025, 0.975)), "months")
```

```
95% Percentile CI: 8.5 17.5 months
```

Bootstrap Distribution of Median

▶ Code



The bootstrap median distribution takes on **few unique values** — this is typical!

Bootstrap t Confidence Interval (Devore 8.5)

Bootstrap t Interval (Simple Version)

If the bootstrap distribution is approximately **normal** with small **bias**, use:

$$\hat{\theta} \pm t_{\alpha/2,n-1} \cdot SE_{boot}$$

where SE_{boot} is the bootstrap standard error.

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If the bootstrap distribution is approximately **normal** with small **bias**, use:

$$\hat{\theta} \pm t_{\alpha/2,n-1} \cdot SE_{boot}$$

where SE_{boot} is the bootstrap standard error.

```r
# Bootstrap t interval for trimmed mean salary
xbar_tr <- mean(salaries, trim = 0.10)
se_boot <- sd(boot_trimmed_sal)
t_crit <- qt(0.975, df = length(salaries) - 1)

ci_lower <- xbar_tr - t_crit * se_boot
ci_upper <- xbar_tr + t_crit * se_boot

cat("Bootstrap t 95% CI for trimmed mean salary:\n")
```

```
Bootstrap t 95% CI for trimmed mean salary:
```

```r
cat("($", round(ci_lower, 0), ", $", round(ci_upper, 0), ")", sep = "")
```

```
($82868, $93992)
```

The Studentized Bootstrap t Interval (Chihara 7.5)

Bootstrap t Interval (More Sophisticated)

For each bootstrap sample, compute:

$$T^* = \frac{\bar{X}^* - \bar{x}}{S^*/\sqrt{n}}$$

Use quantiles $Q^*_{\alpha/2}$ and $Q^*_{1-\alpha/2}$ of the bootstrap t distribution:

$$\left(\bar{x} - Q^*_{1-\alpha/2} \cdot \frac{s}{\sqrt{n}},\ \bar{x} - Q^*_{\alpha/2} \cdot \frac{s}{\sqrt{n}}\right)$$

The Studentized Bootstrap t Interval (Chihara 7.5)

Bootstrap t Interval (More Sophisticated)

For each bootstrap sample, compute:

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Use quantiles $Q^*_{\alpha/2}$ and $Q^*_{1-\alpha/2}$ of the bootstrap t distribution:

$$\left(\bar{x} - Q^*_{1-\alpha/2} \cdot \frac{s}{\sqrt{n}},\ \bar{x} - Q^*_{\alpha/2} \cdot \frac{s}{\sqrt{n}}\right)$$

This method adapts to **skewness** in the data!

Bootstrap t Interval: R Implementation

```r
bootstrap_t_interval <- function(data, B = 10000, conf = 0.95) {
  n <- length(data)
  xbar <- mean(data)
  se <- sd(data) / sqrt(n)

  # Compute bootstrap t statistics
  t_star <- replicate(B, {
    boot_samp <- sample(data, n, replace = TRUE)
    (mean(boot_samp) - xbar) / (sd(boot_samp) / sqrt(n))
  })

  # Quantiles
  alpha <- 1 - conf
  q_lower <- quantile(t_star, alpha/2)
  q_upper <- quantile(t_star, 1 - alpha/2)

  # CI (note the reversal!)
  c(xbar - q_upper * se, xbar - q_lower * se)
}

# Apply to blood pressure data
boot_t_ci <- bootstrap_t_interval(bp_reduction)
cat("Bootstrap t CI:", round(boot_t_ci, 2), "mmHg")
```

Why the Reversal in Bootstrap t?

Understanding the Bootstrap t Formula

The percentile interval uses quantiles **directly**: $(\hat{\theta}^*_{\alpha/2}, \hat{\theta}^*_{1-\alpha/2})$

The bootstrap t interval uses quantiles of the **t-statistics** with reversal:

$$\left(\bar{x} - Q^*_{1-\alpha/2} \cdot SE,\ \bar{x} - Q^*_{\alpha/2} \cdot SE\right)$$

The reversal accounts for the fact that if T^* is large, then $\bar{X}^*$ was above $\bar{x}$, so the lower bound for μ should be reduced.

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Understanding the Bootstrap t Formula

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The reversal accounts for the fact that if T^* is large, then $\bar{X}^*$ was above $\bar{x}$, so the lower bound for μ should be reduced.

This is analogous to how the classical t-interval works: large t values suggest the true mean might be lower than $\bar{x}$.

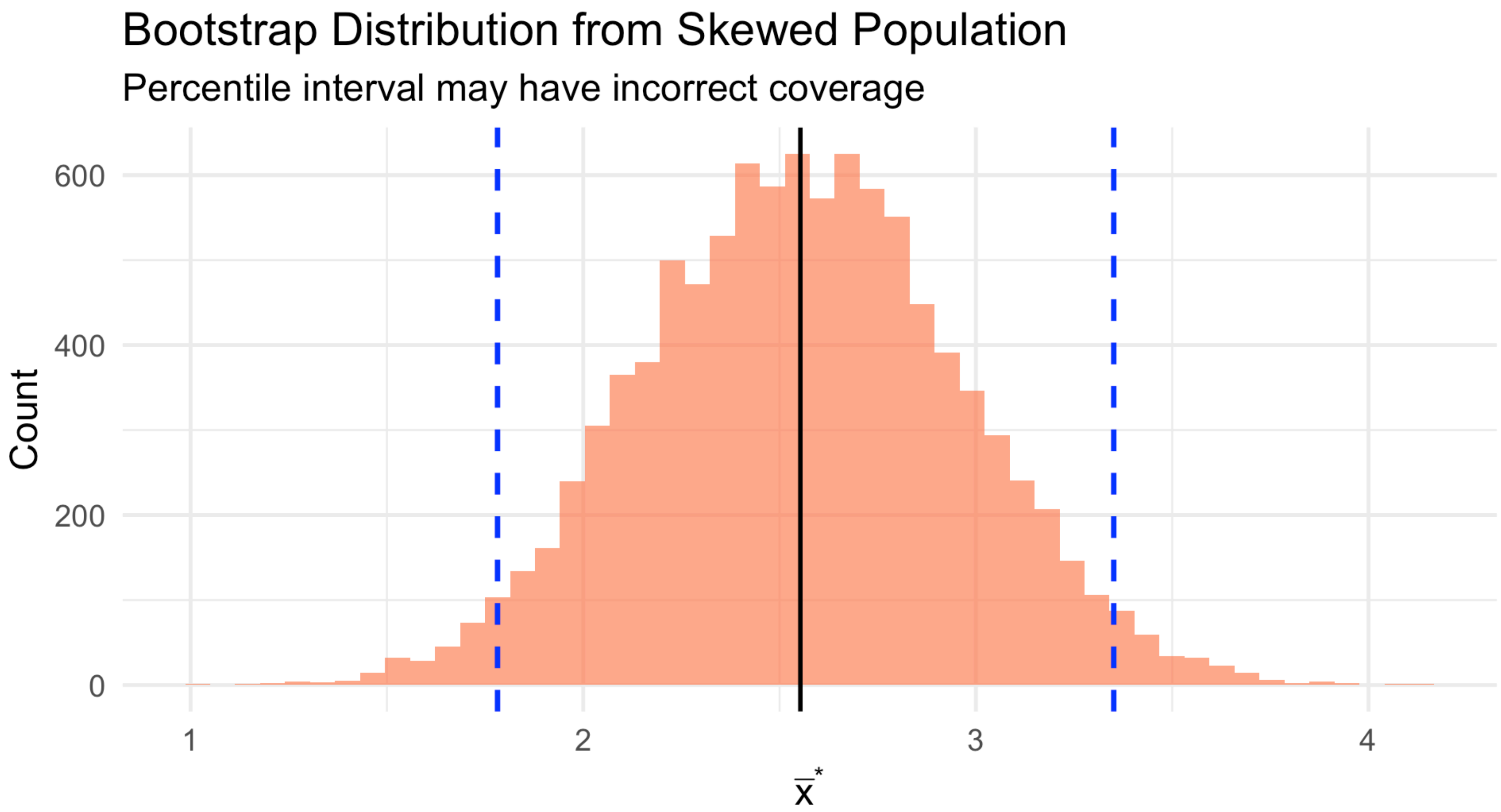
Comparing All Methods

▶ Code

```
# A tibble: 3 × 4
  Method               Lower Upper Width
  <chr>                <dbl> <dbl> <dbl>
1 Classical t           10.5  16.7  6.28
2 Bootstrap Percentile  10.9  16.5  5.53
3 Bootstrap t           10.8  17.2  6.47
```

When Percentile May Be Suboptimal: Skewed Distributions

▶ Code



Practical Considerations

How Many Bootstrap Samples? (Chihara 5.9)

Guidelines

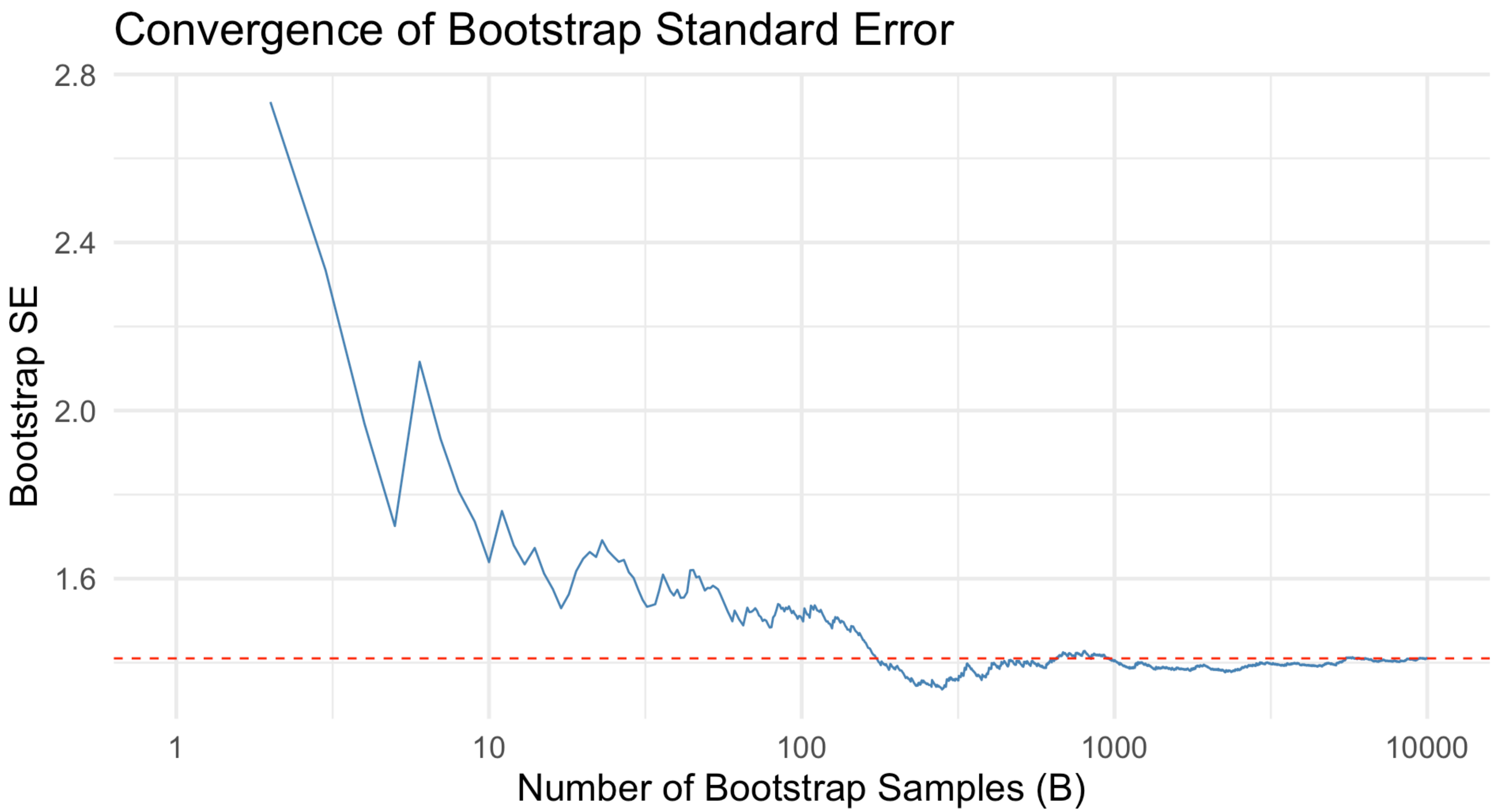
- **Standard errors:** $B = 1,000$ is often sufficient
- **Confidence intervals:** $B = 5,000$ to $10,000$ recommended
- **Percentile tails:** More samples needed for accurate tail quantiles

```r
# Demonstrating convergence of bootstrap SE
set.seed(551)
B_values <- c(100, 500, 1000, 5000, 10000)

convergence <- map_dfr(B_values, function(B) {
  boot_means <- replicate(B, mean(sample(bp_reduction, length(bp_reduction), replace = TRUE
  tibble(B = B, SE = sd(boot_means))
})

convergence
```

```
# A tibble: 5 × 2
      B    SE
  <dbl> <dbl>
1   100  1.51
2   500  1.37
3  1000  1.38
4  5000  1.41
```

Visualizing Bootstrap Convergence

▶ Code



Classical vs. Bootstrap: Guidelines (Chihara 7.5.1)

Situation	Recommendation
Small n (≤10), normal data	Classical t preferred
Small n (≤10), skewed data	Bootstrap t (with caution)
Large n, normal data	Either works well
Large n, skewed data	Bootstrap t preferred
Complex statistic (trimmed mean, ratio)	Bootstrap (no classical alternative)

Classical vs. Bootstrap: Guidelines (Chihara 7.5.1)

Situation	Recommendation
Small n (≤10), normal data	Classical t preferred
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Large n, normal data	Either works well
Large n, skewed data	Bootstrap t preferred
Complex statistic (trimmed mean, ratio)	Bootstrap (no classical alternative)

Key insight: The bootstrap **learns skewness from the sample**, but small samples may not accurately represent population skewness.

When Does Bootstrap Work Well?

Bootstrap Works Well When:

1. The sample is **representative** of the population
2. The sample size is **not too small** ($n \geq 15$ typically)
3. The statistic is **smooth** (means, variances, regression coefficients)
4. The parameter is well-defined and has finite variance

When Does Bootstrap Work Well?

Bootstrap Works Well When:

1. The sample is **representative** of the population
2. The sample size is **not too small** ($n \geq 15$ typically)
3. The statistic is **smooth** (means, variances, regression coefficients)
4. The parameter is well-defined and has finite variance

Bootstrap May Struggle When:

1. Sample is very small ($n < 10$)
2. Data has extreme outliers not representative of population
3. Statistic is very discrete (e.g., median with few unique values)
4. Estimating extreme quantiles
5. Population has infinite variance (Cauchy, Pareto with $\alpha \leq 2$)

Medical Applications

Medical Application: Relative Risk (Chihara 5.5)

In a cardiovascular study:

- High BP group: 55 events in 3,338 men
- Low BP group: 21 events in 2,676 men

```r
# Relative risk
p1 <- 55/3338  # High BP group
p2 <- 21/2676  # Low BP group
rr_hat <- p1/p2

cat("Risk (high BP):", round(p1 * 100, 2), "%\n")
```

```
Risk (high BP): 1.65 %
```

```r
cat("Risk (low BP):", round(p2 * 100, 2), "%\n")
```

```
Risk (low BP): 0.78 %
```

```r
cat("Relative Risk:", round(rr_hat, 2))
```

```
Relative Risk: 2.1
```

Bootstrap for Relative Risk

```r
set.seed(551)
n1 <- 3338; x1 <- 55
n2 <- 2676; x2 <- 21

# Bootstrap
boot_rr <- replicate(10000, {
  # Resample from each group (binary outcomes)
  boot_p1 <- sum(sample(c(rep(1, x1), rep(0, n1 - x1)), n1, replace = TRUE)) / n1
  boot_p2 <- sum(sample(c(rep(1, x2), rep(0, n2 - x2)), n2, replace = TRUE)) / n2

  # Handle division by zero
  if (boot_p2 == 0) return(NA)
  boot_p1 / boot_p2
})

boot_rr <- boot_rr[!is.na(boot_rr)]

cat("Bootstrap SE:", round(sd(boot_rr), 3), "\n")
```

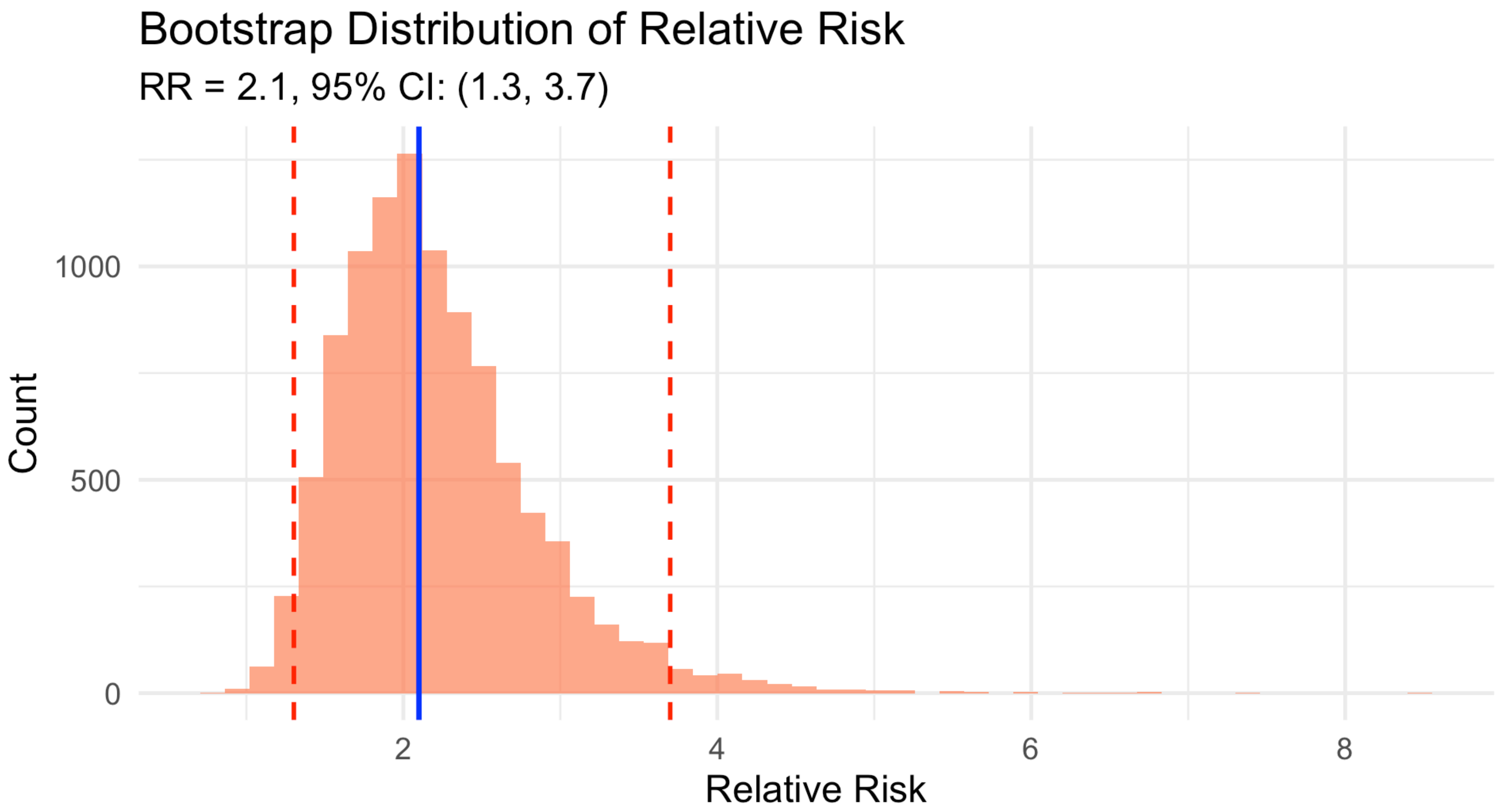
```
Bootstrap SE: 0.627
```

```r
cat("95% Percentile CI:", round(quantile(boot_rr, c(0.025, 0.975)), 2))
```

```
95% Percentile CI: 1.3 3.7
```

Relative Risk: Bootstrap Distribution

▶ Code



Interpretation: High BP patients have 1.3 to 3.7 times the cardiovascular risk (95% CI).

Clinical Trial: Comparing Treatments

A trial compares time to pain relief for two analgesics:

```r
set.seed(551)
# Treatment A: faster, less variable
drug_A <- c(12, 18, 15, 22, 14, 16, 20, 13, 17, 19)
# Treatment B: slower, more variable
drug_B <- c(25, 18, 35, 22, 28, 15, 42, 20, 30, 24)

cat("Drug A - Mean:", round(mean(drug_A), 1), ", SD:", round(sd(drug_A), 1), "\n")
```

```
Drug A - Mean: 16.6 , SD: 3.2
```

```r
cat("Drug B - Mean:", round(mean(drug_B), 1), ", SD:", round(sd(drug_B), 1), "\n")
```

```
Drug B - Mean: 25.9 , SD: 8.2
```

```r
cat("Difference (B - A):", round(mean(drug_B) - mean(drug_A), 1), "minutes")
```

```
Difference (B - A): 9.3 minutes
```

Two-Sample Bootstrap

```r
set.seed(551)
# Bootstrap the difference in means
boot_diff <- replicate(10000, {
  boot_A <- sample(drug_A, length(drug_A), replace = TRUE)
  boot_B <- sample(drug_B, length(drug_B), replace = TRUE)
  mean(boot_B) - mean(boot_A)
})

# Results
diff_ci <- quantile(boot_diff, c(0.025, 0.975))
cat("Bootstrap SE:", round(sd(boot_diff), 2), "\n")
```

```
Bootstrap SE: 2.62
```

```r
cat("95% CI for difference:", round(diff_ci, 2), "minutes")
```

```
95% CI for difference: 4.5 14.7 minutes
```

Two-Sample Bootstrap

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set.seed(551)
# Bootstrap the difference in means
boot_diff <- replicate(10000, {
  boot_A <- sample(drug_A, length(drug_A), replace = TRUE)
  boot_B <- sample(drug_B, length(drug_B), replace = TRUE)
  mean(boot_B) - mean(boot_A)
})

# Results
diff_ci <- quantile(boot_diff, c(0.025, 0.975))
cat("Bootstrap SE:", round(sd(boot_diff), 2), "\n")
```

```
Bootstrap SE: 2.62
```

```r
cat("95% CI for difference:", round(diff_ci, 2), "minutes")
```

```
95% CI for difference: 4.5 14.7 minutes
```

Since CI excludes 0, there's evidence Drug B takes longer to provide relief.

Bootstrap for Odds Ratio

In a case-control study of lung cancer and smoking:

```r
# Lung cancer cases
cases_smokers <- 85; cases_nonsmokers <- 15
# Controls
controls_smokers <- 60; controls_nonsmokers <- 40

# Sample odds ratio
or_hat <- (cases_smokers * controls_nonsmokers) / (cases_nonsmokers * controls_smokers)
cat("Sample Odds Ratio:", round(or_hat, 2))
```

```
Sample Odds Ratio: 3.78
```

Bootstrap for Odds Ratio (cont.)

```
set.seed(551)
# Bootstrap
n_cases <- cases_smokers + cases_nonsmokers
n_controls <- controls_smokers + controls_nonsmokers

boot_or <- replicate(10000, {
  # Resample cases
  boot_cases <- sample(c(rep(1, cases_smokers), rep(0, cases_nonsmokers)),
                       n_cases, replace = TRUE)
  boot_case_smoke <- sum(boot_cases)
  boot_case_nonsmoke <- n_cases - boot_case_smoke

  # Resample controls
  boot_controls <- sample(c(rep(1, controls_smokers), rep(0, controls_nonsmokers)),
                          n_controls, replace = TRUE)
  boot_ctrl_smoke <- sum(boot_controls)
  boot_ctrl_nonsmoke <- n_controls - boot_ctrl_smoke

  # OR (with continuity correction)
  (boot_case_smoke + 0.5) * (boot_ctrl_nonsmoke + 0.5) /
    ((boot_case_nonsmoke + 0.5) * (boot_ctrl_smoke + 0.5))
})
```

Summary and Looking Ahead

Bootstrap CIs Summary

Method	Formula	When to Use
Percentile	$(\hat{\theta}^*_{\alpha/2}, \hat{\theta}^*_{1-\alpha/2})$	Symmetric bootstrap dist., low bias
Bootstrap t (simple)	$\hat{\theta} \pm t_{\alpha/2} \cdot SE_{boot}$	Normal bootstrap dist.
Bootstrap t (studentized)	Uses bootstrap t-statistics	Skewed data, more accurate

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Key Point

Bootstrap t intervals have better coverage than percentile intervals for skewed data, but require the bootstrap SE to be meaningful.

What the Bootstrap Can and Cannot Do (Chihara 5.2.2)

Can estimate:

- Standard error
- Bias
- Skewness
- Confidence intervals

Cannot improve:

- Point estimates (still centered at $\hat{\theta}$, not θ)
- A bad/unrepresentative sample

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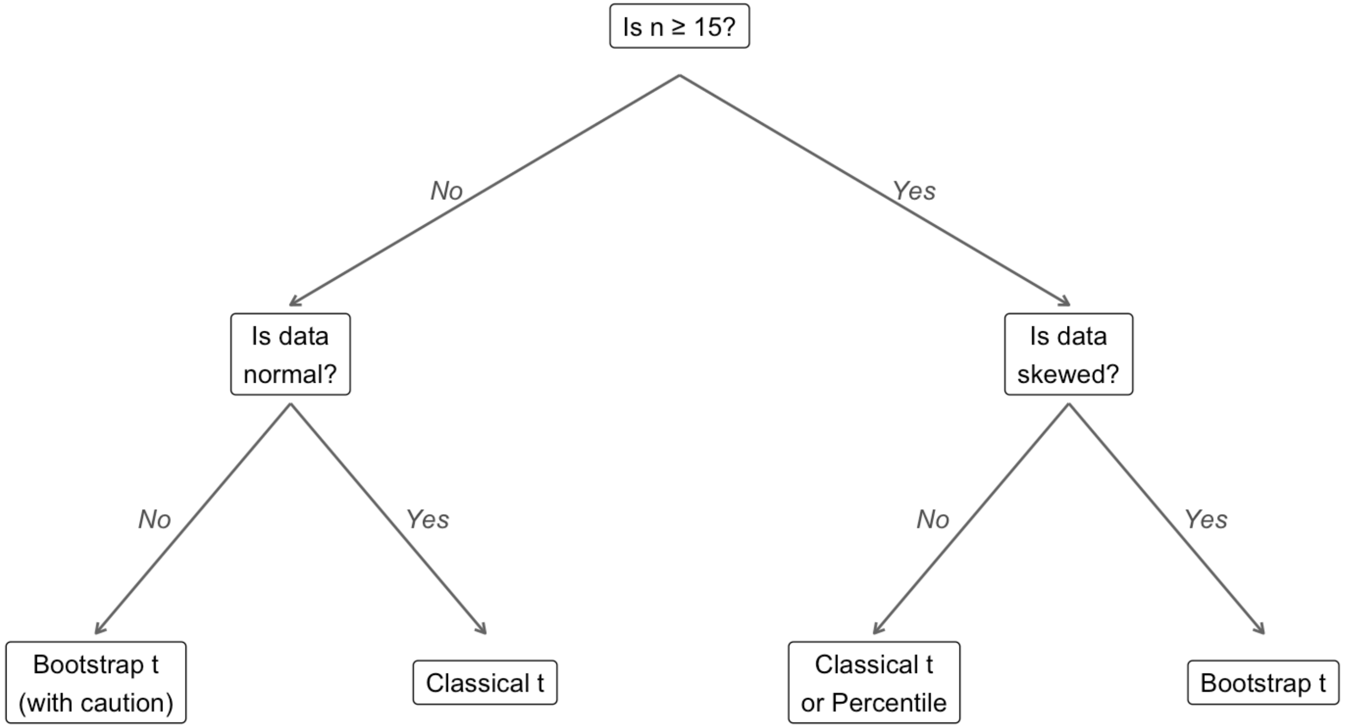
⚠ **Important Reminder**

The bootstrap does **not create new information**. It uses the sample you have to understand the variability of your estimate.

Choosing a Method: Decision Tree

Choosing a Confidence Interval Method



Looking Ahead

Next Topics:

- **Hypothesis testing framework** (Lesson 11)
- Type I and Type II errors
- P-values and their interpretation
- Power and sample size

Looking Ahead

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- **Hypothesis testing framework** (Lesson 11)
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Later:

- Two-sample inference
- Permutation tests
- Bootstrap hypothesis testing

References

- Devore, Berk, and Carlton. *Modern Mathematical Statistics with Applications* (Springer). Section 8.5
- Chihara and Hesterberg. *Mathematical Statistics with Resampling and R* (Wiley). Chapters 5, 7
- Efron, B. and Tibshirani, R. (1993). *An Introduction to the Bootstrap*. Chapman & Hall.

Questions?

Thank you!